23. Name the seven contrasting traits of Mendel.
24. What is meant by true breeding or pure breeding lines / strain?
25. Give the names of the scientists who rediscovered Mendelism.
26. What is back cross?
27. Define Genetics.
28. What are multiple alleles
29. What are the reasons for Mendel's successes in his breeding experiment?
30. Explain the law of dominance in monohybrid cross.
31. Differentiate incomplete dominance and codominance.
32. What is meant by cytoplasmic inheritance
33. Describe dominant epistasis with an example.
34. Explain polygenic inheritance with an example.
35. Differentiate continuous variation with discontinuous variation.
36. Explain with an example how single genes affect multiple traits and alleles the phenotype of an organism.
37. Bring out the inheritance of chloroplast gene with an example.

Glossary

Alleles: Alternative forms of a gene.

Back Cross: Crosses between F_1 off-springs with either of the two parents (hybrid) are known as back cross

F_1 / First Filial Generation: The second stage of Mendel's experiment is called F_1 generation

Gene: The determinant of a characteristic of an organism (Mendelian factor). Gene symbols are underlined or italicized.

Genetic Code: The set of 64 triplets of bases (codons) corresponding to the twenty amino acids in proteins and the signals for initiation and termination of polypeptide synthesis.

Genotype: The types of alleles in a single individual is called genotype

Genome: The total complement of genes contained in a cell.

Heterozygous: Diploid organisms that have two different allels at a specific gene locus are said to be heterozygous.

Homozygous: A diploid organism in which both alleles are the same at a given gene locus is said to be homozygous.

Hybrid Vigour or Heterosis: The superiority of hybrid over either of its parents in one or more traits.

Locus: The site or position of a particular gene on a chromosome.

Phenotype: The physical expression of an individuals gene. The physical observable characteristics of an organism.

Punnett Square / Checkerboard: A sort of cross-multiplication matrix used in the prediction of the outcome of a genetic cross, in which male and female gametes and their frequencies are arranged along the edges.

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